

Asymptotically Good Quantum LDPC Codes

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Published in: IEEE Transactions on Information Theory (2021) | Archival ID: QV-QC-90

1. ABSTRACT & THEORETICAL FOUNDATION

We prove the long-standing conjecture of the existence of asymptotically good quantum LDPC codes with simultaneously constant encoding rate $k/n = \Omega(1)$ and linear code distance $d/n = \Omega(1)$.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Surface codes need 1,000 physical qubits to protect 1 logical qubit. qLDPC codes use clever high-dimensional geometry to protect 100 logical qubits with only 1,500 physical qubits!"

3. Mathematical Formulation & Unitary Rule

$$\lim_{n \rightarrow \infty} \frac{k}{n} > 0, \quad \lim_{n \rightarrow \infty} \frac{d}{n} > 0, \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{d}{k} > 0$$

Asymptotically good qLDPC codes maintain non-zero logical rate k/n and relative distance d/n as system size scales.

4. Step-by-Step Circuit Sequence

Bipartite Tanner graph stabilizer check schedule utilizing long-range optical tweezer shuttling or 3D routing.

5. Executable IBM Qiskit (Python) Code

```
# [[144, 12, 12]] Bivariate Bicycle qLDPC Code
n_phys, k_logical, distance = 144, 12, 12
ratio = k_logical / n_phys
print(f"qLDPC Encoding efficiency: {ratio:.2%} ({k_logical} logicals in {n_phys} physicals)")
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

$O(n)$ linear-time belief propagation and ordered statistics decoding (OSD) for large-scale FTQC

7. Key Applications & Future Research Directions

- Neutral atom quantum computing architectures (QuEra, Harvard, MIT demonstrations)
- High-density quantum memory storage modules
- Fault-tolerant quantum interconnects

Future Work:

Hardware-adapted 3D routing for neutral-atom optical tweezers and low-latency Belief Propagation decoders.